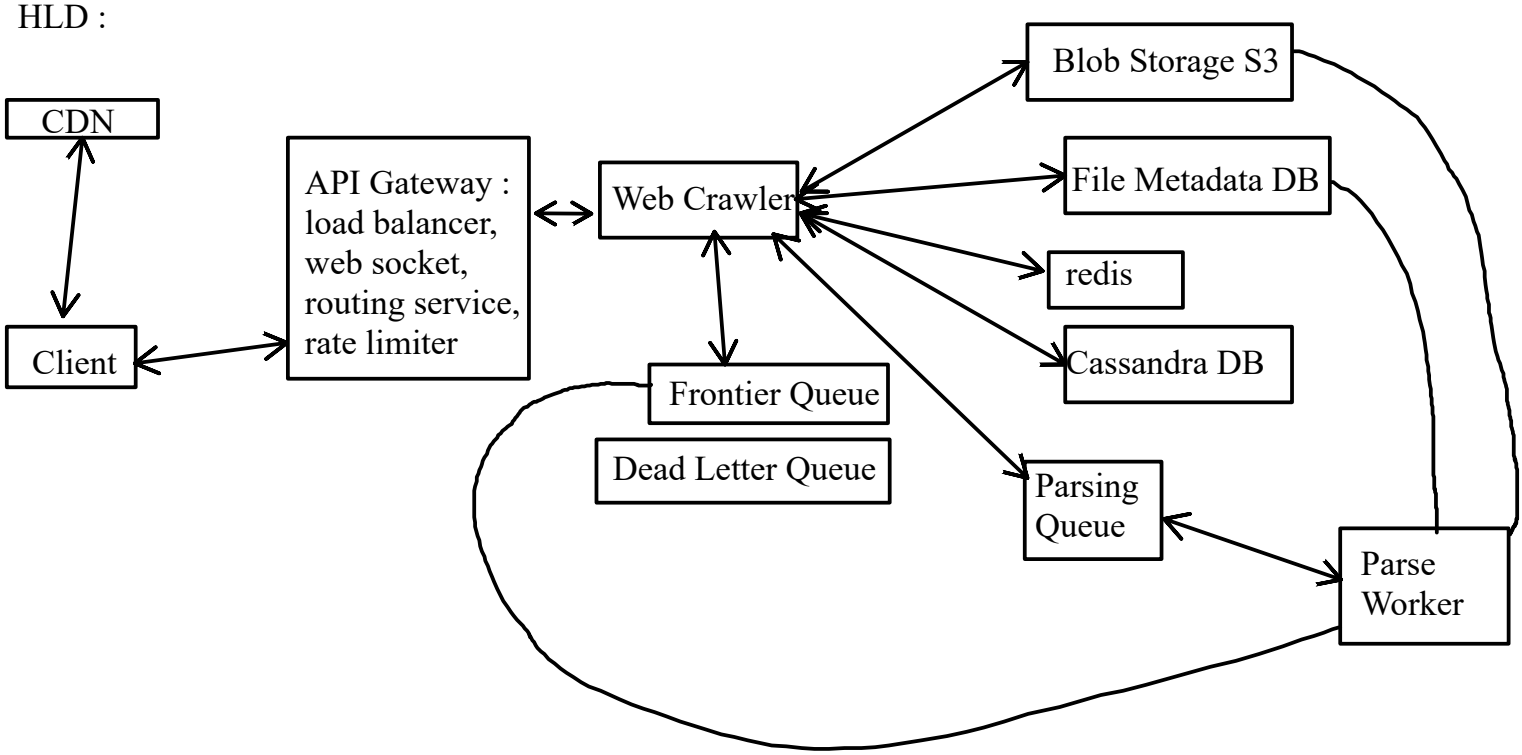
WebCrawler

Functional requirements :
user should be able to download file from fileMetadata,
user should be able to place downloads for miltiple files as well

Non-functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency ,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
url,
filemetadata

API :
POST:/download
{
userID,
{filemetadata}
}
GET:/search/{fileID}

HLD :